EC5.101 - Assignment 1

1. a) $x(t) = t^2 \cos(3t)$

$\Rightarrow x(-t) = (-t)^2 \cos(-3t) = t^2 \cos(3t) = x(t)$

$\therefore$ EVEN SIGNAL.

b) $x(t) = t\cos(70t) + t\sqrt{1+t^2}$

$\Rightarrow x(-t) = -t\cos(-70t) - t\sqrt{1+(-t)^2}$

$= -\left[t\cos(70t) + t\sqrt{1+t^2}\right] = -x(t)$

$\therefore$ ODD SIGNAL

c) $x(t) = \sin^2(7t) + \sin(t)$

$\Rightarrow x(-t) = \sin^2(-7t) + \sin(-t)$

$= \sin^2(7t) - \sin(t) \neq x(t), x(-t)$

$\therefore$ NEITHER EVEN NOR ODD

Now, $T = LCM\left(\frac{\pi}{7}, 2\pi\right) = 2\pi$.

$\Rightarrow x(t + T/2) = \cancel{\sin^2(7+\pi)t)}\ \sin^2(7t+7\pi) + \sin(t+\pi)$

$= \sin^2(7t) - \sin(t) \neq -x(t)$

$\therefore$ THE SIGNAL DOES NOT SHOW HALF-WAVE SYM.

d) $x(t) = \log(\cos(t))$

$\Rightarrow x(-t) = \log(\cos(-t)) = \log(\cos(t)) = x(t)$

$\therefore$ EVEN SIGNAL

Now, $T = 2\pi$.

$\Rightarrow x(t+\pi) = \log(\cos(\pi+t)) = \log(-\cos(t)) \rightarrow$ UNDEFINED

$\therefore$ HALF-WAVE SYM. DOES NOT EXIST.

2. a) $x_1(t) = 1 + \cos(\omega_0 t) + \cancel{3}\cos(3\omega_0 t) + 2\sin(\omega_0 t)$

$x_2(t) = -\cos(\omega_0 t) - \cos(3\omega_0 t) - 2\sin(\omega_0 t)$

Now, since in $\int_{\langle T \rangle} x_1(t)\, x_2(t)\, dt$, terms like $\int_{\langle T \rangle} \cos^2(\omega_0 t)\, dt$ appear, $\langle x_1(t), x_2(t) \rangle \neq 0$

$\therefore$ $x_1(t)$ & $x_2(t)$ are <u>not orthogonal</u>.

b) $x_1(t) = \cos(\omega_0 t) + \cos(3\omega_0 t) + 2\sin(2\omega_0 t)$

$x_2(t) = 1 + \cos(2\omega_0 t) + 2\sin(\omega_0 t)$

Here, the product $x_1(t) \cdot x_2(t)$ only has singular sinusodial terms or of the form $\cos(k_1 \omega_0 t)\cos(k_2 \omega_0 t)$ and likewise, which all integrate to 0 over T.

$\Rightarrow \langle x_1(t), x_2(t) \rangle = 0.$

$\therefore$ $x_1(t)$ & $x_2(t)$ <u>are orthogonal</u>.

3. $x(t) = \sin(2t) + \sin^2(t) = \sin(2t) + \dfrac{1 - \cos(2t)}{2}$

a) $$a_0 = \frac{1}{\pi} \int_0^{\pi} \frac{2\sin(2t) + 1 - \cos(2t)}{2}\, dt$$

$$= \frac{1}{2\pi} \left[\frac{-2\cos(2t)}{2} + t - \frac{\sin(2t)}{2} \right]_0^{\pi}$$

$$= \frac{1}{2\pi} \left[-1 + \pi - (-1) \right] = \frac{1}{2}$$

$$x(t) = \frac{1}{2} - \frac{1}{2}\cos(2t) + \sin(2t)$$

$$\therefore\ x(t) \xleftrightarrow{FS} \{ \underbrace{\tfrac{1}{2}}_{a_0}, [\underbrace{-\tfrac{1}{2}}_{a_1}, 0, \ldots], [\underbrace{1}_{b_1}, 0, \ldots] \}$$

b) $x(t) \xleftrightarrow{FS} \{\frac{1}{2}d_0, [d_1, \ldots], [d_{-1}, \ldots]\}$

$d_0 = a_0 = 1/2$; $d_1 = \frac{a_1 - jb_1}{2}$; $d_{-1} = \frac{a_1 + jb_1}{2}$

$\Rightarrow d_1 = \frac{-\frac{1}{2} - j}{2} = \frac{-1-2j}{4}$; $d_{-1} = \frac{-\frac{1}{2}+j}{2} = \frac{-1+2j}{4}$

~~4. a) $2x(2t)$ $x(t)$ $\longleftrightarrow$~~

4. $x(t) \xleftrightarrow{FS} \{j, -1, 2, \underset{\uparrow}{0}, 2, -1, -j\}$

a) $2x(2t) \xleftrightarrow{FS} \{2j, -2, 4, 0, 4, -2, -2j\}$
($\because$ FCs are independent of signal frequency)

b) $x(2-t) \xleftrightarrow{FS} \{-je^{-6j\omega_0}, -e^{-4j\omega_0}, 2e^{-2j\omega_0}, 0, 2e^{2j\omega_0}, -e^{4j\omega_0}, je^{6j\omega_0}\}$

$$\left[\begin{aligned} \because d_k e^{jk\omega_0(2-t)} &= \underbrace{d_k e^{2jk\omega_0}}\cdot e^{-jk\omega_0 t} \\ &= d_K e^{jK\omega_0 t}, \quad K = -k \end{aligned}\right]$$

(assuming signal frequency ω_0)

c) $x(t) - x(-t) \xleftrightarrow{FS} \{2j, 0, 0, 0, 0, 0, -2j\}$
($\because x(-t) \xleftrightarrow{FS} \{-j, -1, 2, 0, 2, -1, j\}$)

d) $2 - x(t) \xleftrightarrow{FS} \{-j, +1, -2, 2, -2, +1, j\}$
($\because$ constant term appears as d_0)

5. $\int_0^1 |x(t)|^2 dt = 4 = |a_{-1}|^2 + |a_0|^2 + |a_1|^2$

a) For even signals, $Im(x(t)) = 0$.

$\Rightarrow$ ~~a_{-1}~~ $Im(a_{-1} \pm a_1)$

$\Rightarrow Im(a_{-1} + a_1) = 0$

$\Rightarrow Im(j + a_1) = 0 \quad \Rightarrow \underline{a_1 = -j}$

$\therefore |j|^2 + |a_0|^2 + |-j|^2 = 4$

$\Rightarrow |a_0|^2 = 2 \quad \Rightarrow |a_0| = \sqrt{2}$

$\therefore \underline{\underline{a_0 = \pm\sqrt{2}}}$

b) Since $x(t)$ is even, $a_1 = a_{-1}^*$ (real-valued)
Also, since it has half-wave symmetry, $a_0 = 0$

$\therefore |a_1|^2 + |a_{-1}|^2 + 0 = 4$

$\Rightarrow |a_{-1}^*|^2 + |a_{-1}|^2 = 4$

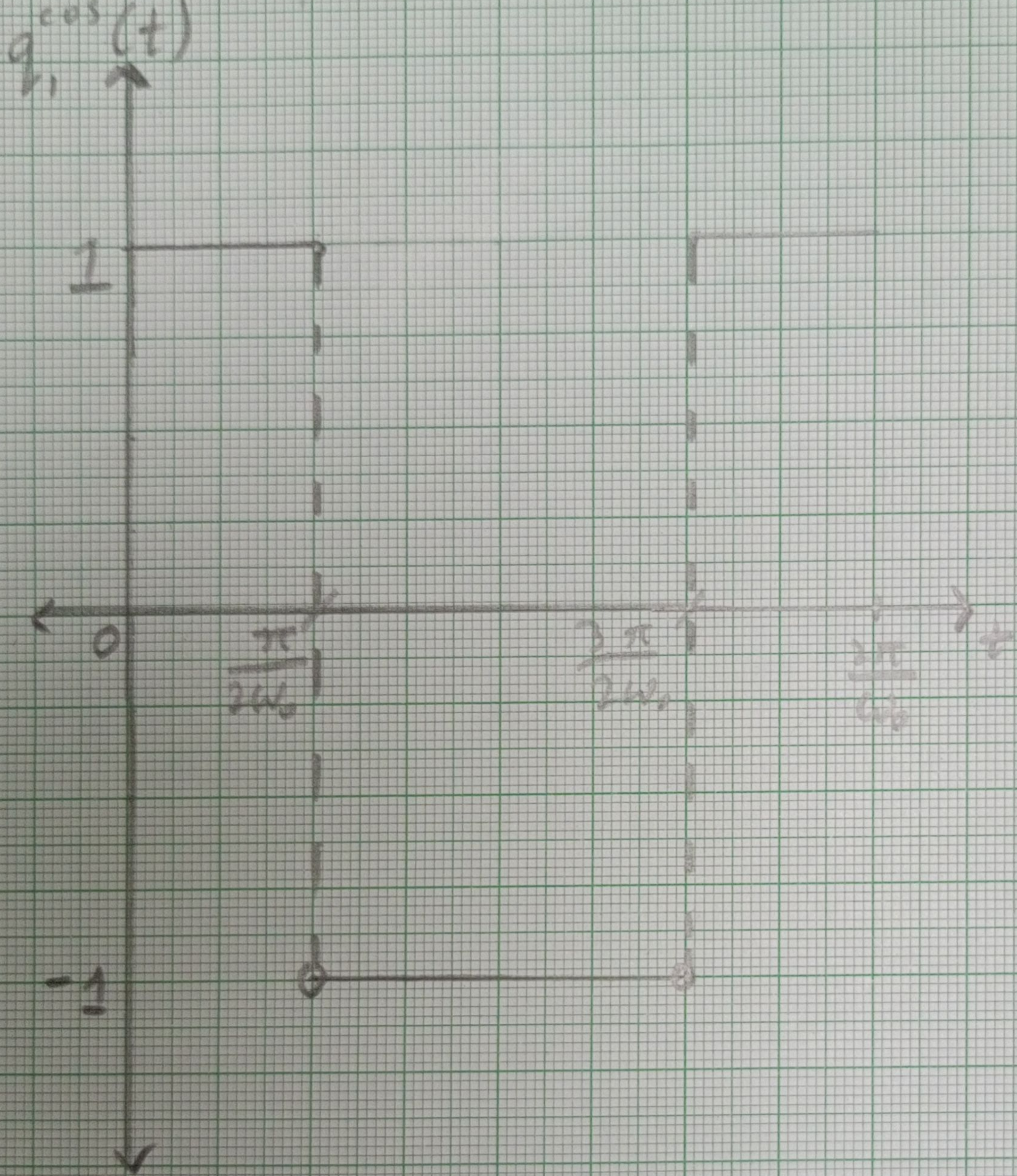
$\Rightarrow 2|a_{-1}|^2 = 4 \Rightarrow |a_{-1}|^2 = 2$

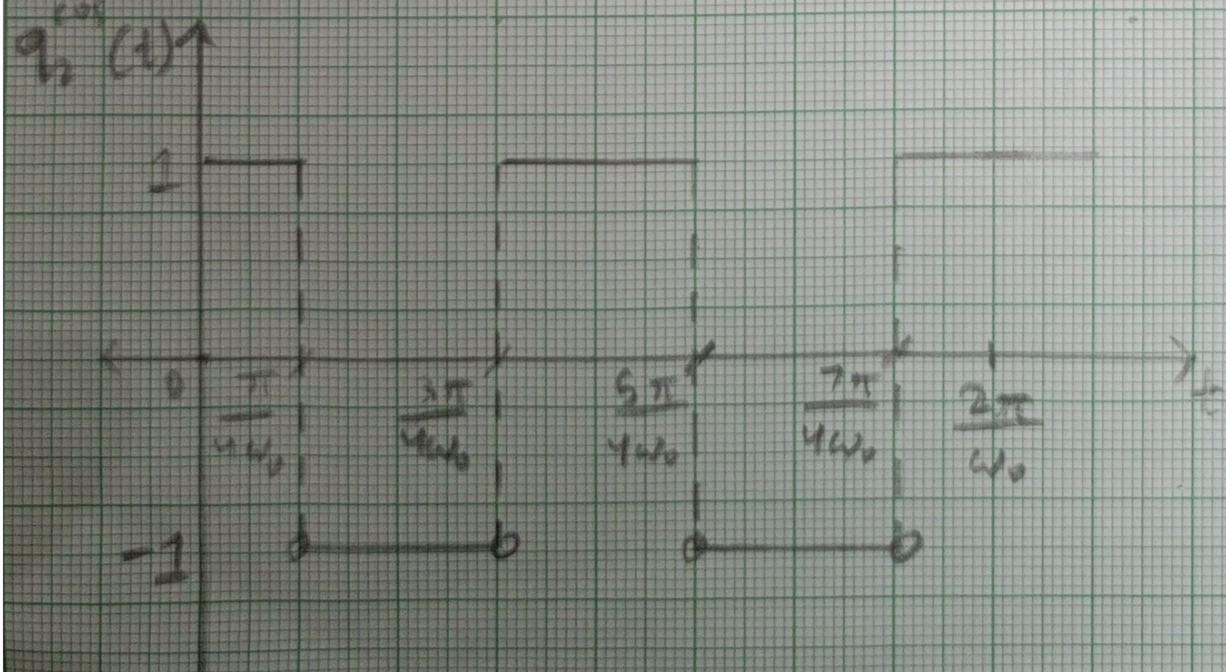
$\Rightarrow |a_{-1}| = \sqrt{2} \qquad \therefore \underline{a_{-1} = \pm\sqrt{2}}$

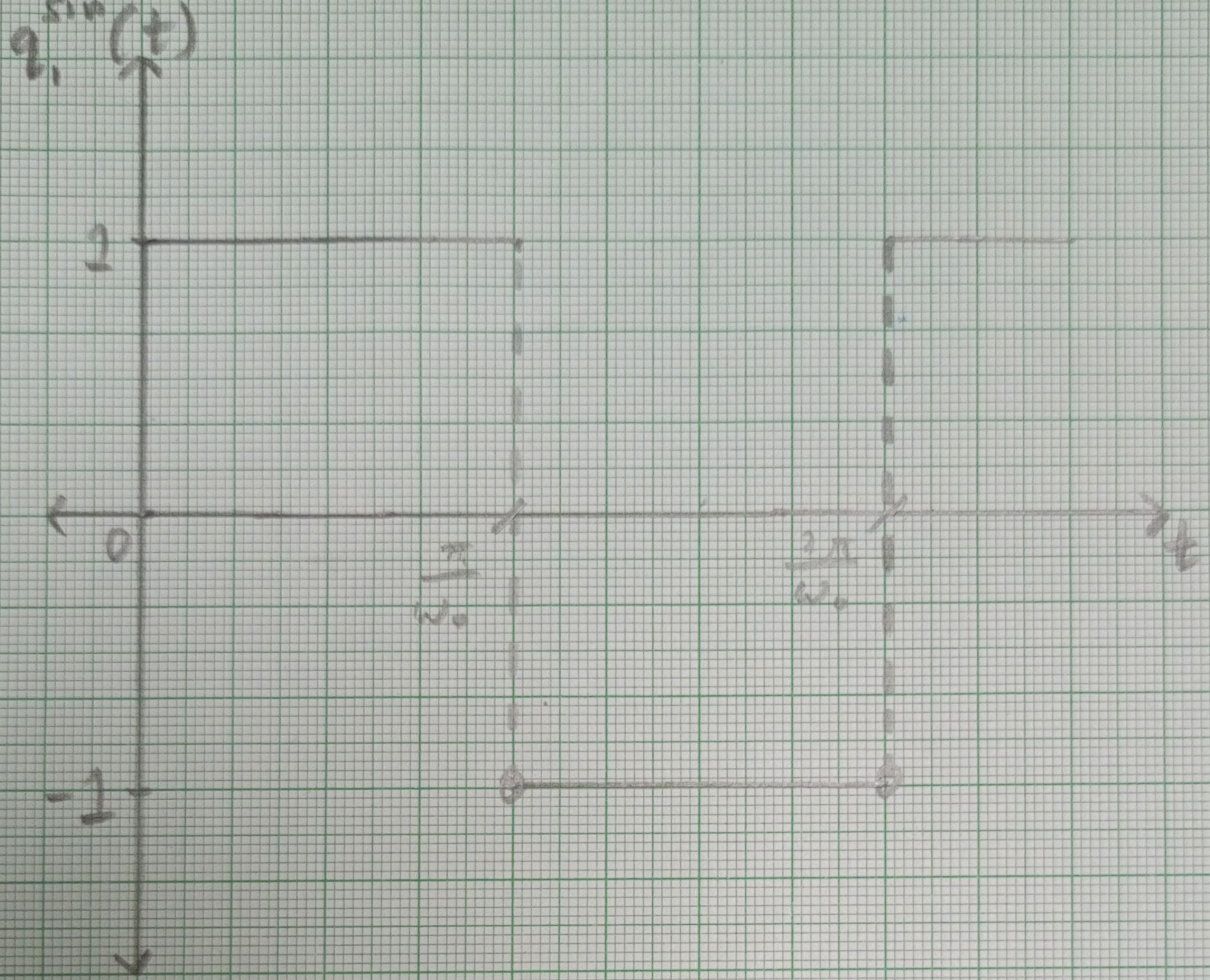
6.a) $q_1^{cos}(t) = sgn(\cos(\omega_0 t))$; $q_2^{cos}(t) = sgn(\cos(2\omega_0 t))$

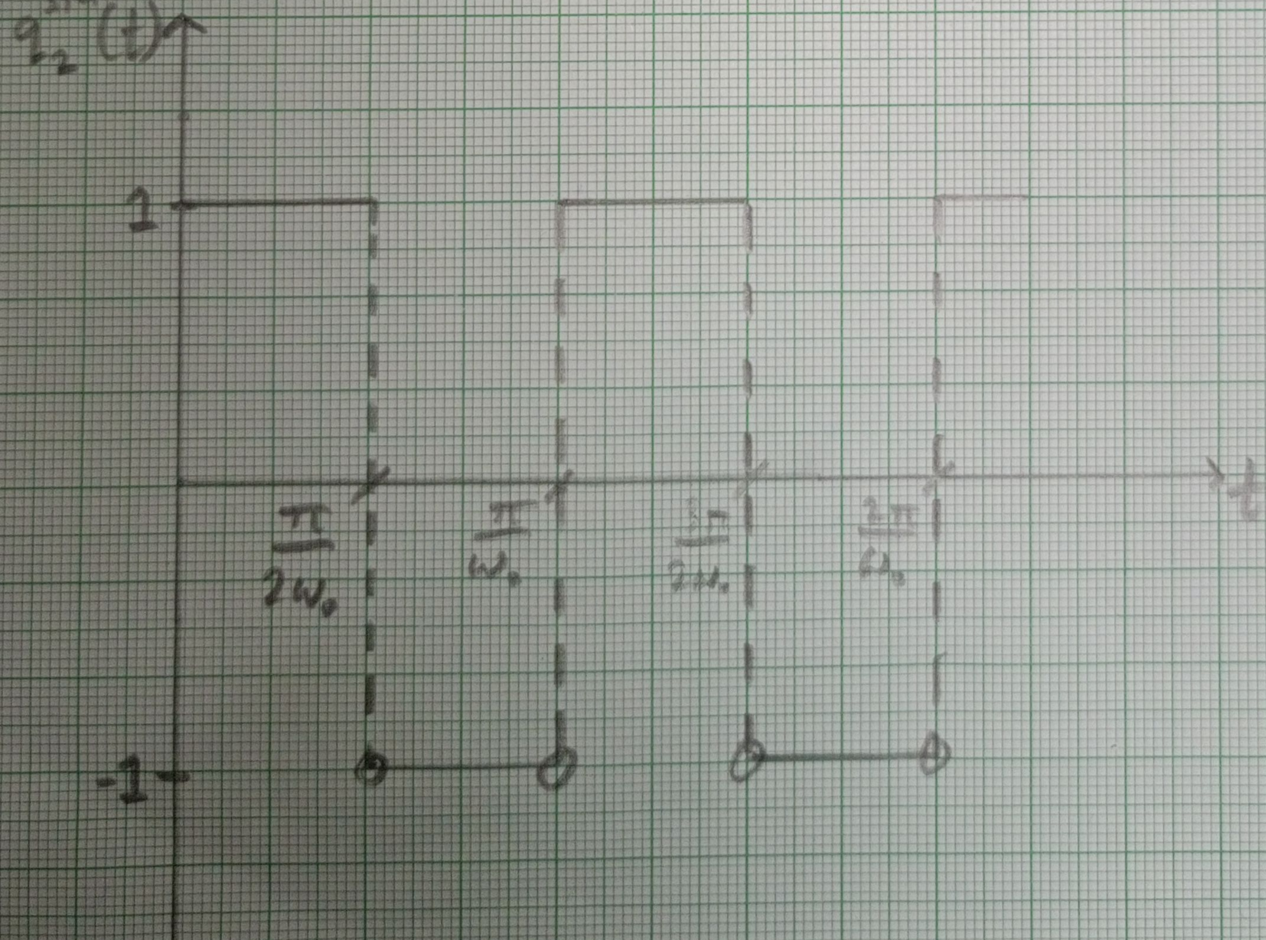
$q_1^{sin}(t) = sgn(\sin(\omega_0 t))$; $q_2^{sin}(t) = sgn(\sin(2\omega_0 t))$

where
[~~For~~ $q_k^{cos/sin}(t) = 1$ for $cos/sin(k\omega_0 t) = 0$]


$q_1^{cos}(t)$
1
0
-1
$\frac{\pi}{2\omega_0}$
$\frac{3\pi}{2\omega_0}$
$\frac{2\pi}{\omega_0}$
t


$q_2^{cos}(t)$
1
0
-1
$\frac{\pi}{4\omega_0}$
$\frac{3\pi}{4\omega_0}$
$\frac{5\pi}{4\omega_0}$
$\frac{7\pi}{4\omega_0}$
$\frac{2\pi}{\omega_0}$
t


$q_1^{sin}(t)$
1
0
$\frac{\pi}{\omega_0}$
$\frac{2\pi}{\omega_0}$
t
-1


$q_2^{sin}(t)$
1
$\frac{\pi}{2\omega_0}$
$\frac{\pi}{\omega_0}$
$\frac{3\pi}{2\omega_0}$
$\frac{2\pi}{\omega_0}$
t
-1

b) For $q_{k_1}(t)$ & $q_{k_2}(t)$, the time period,

$$T = LCM\left(\frac{2\pi}{k_1\omega_0}, \frac{2\pi}{k_2\omega_0}\right)$$

• $\boxed{q_1^{cos}(t) \; \& \; q_2^{cos}(t)}$ are orthogonal over $[0,T]$;

$$\langle q_1^{cos}(t), q_2^{cos}(t) \rangle = \int_0^{2\pi/\omega_0} q_1^{cos}(t)\, q_2^{cos}(t)\, dt$$

$$= \int_0^{\pi/4\omega_0} 1 \cdot 1 \, dt + \int_{\pi/4\omega_0}^{\pi/2\omega_0} 1(-1)\, dt + \int_{\pi/2\omega_0}^{3\pi/4\omega_0} (-1)\cdot(-1)\, dt$$

$$+ \int_{3\pi/4\omega_0}^{5\pi/4\omega_0} (-1)\cdot 1 \, dt + \int_{5\pi/4\omega_0}^{3\pi/2\omega_0} (-1)(-1)\, dt + \int_{3\pi/2\omega_0}^{7\pi/4\omega_0} (1)(-1)\, dt$$

$$+ \int_{7\pi/4\omega_0}^{2\pi/\omega_0} 1 \cdot 1 \, dt = \frac{\pi}{\omega_0}\left[\frac{1}{4} - \frac{1}{4} + \frac{1}{4} - \frac{1}{2} + \frac{1}{4} - \frac{1}{4} + \frac{1}{4}\right]$$

$$= \underline{\underline{0}}$$

• $\boxed{q_1^{sin}(t) \; \& \; q_2^{sin}(t)}$ are orthogonal over $[0,T]$;

$$\langle q_1^{sin}(t), q_2^{sin}(t) \rangle = \int_0^{2\pi/\omega_0} q_1^{sin}(t)\, q_2^{sin}(t)\, dt$$

$$= \frac{\pi}{\omega_0}\left[\frac{1}{2} - \frac{1}{2} - \frac{1}{2} + \frac{1}{2}\right] = \underline{\underline{0}}$$

(Proceeding similarly as before)

- $\boxed{q_1^{cos}(t) \,\&\, q_1^{sin}(t)}$ are orthogonal over $[0,T]$;

$$<q_1^{cos}(t), q_1^{sin}(t)> = \int_0^{2\pi/\omega_0} q_1^{cos}(t)\, q_1^{sin}(t)\, dt$$

$$= \int_0^{\pi/2\omega_0} 1 \cdot 1\, dt + \int_{\pi/2\omega_0}^{\pi/\omega_0} (-1)\cdot 1\, dt + \int_{\pi/\omega_0}^{3\pi/2\omega_0} (-1)(-1)\, dt$$

$$+ \int_{3\pi/2\omega_0}^{2\pi/\omega_0} 1(-1)\, dt = \frac{\pi}{\omega_0}\left[\frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2}\right] = \underline{0}$$

- $\boxed{q_2^{cos}(t) \,\&\, q_2^{sin}(t)}$ are orthogonal over $[0,T]$;

$$<q_2^{cos}(t), q_2^{sin}(t)> = \int_0^{2\pi/\omega_0} q_2^{cos}(t)\, q_2^{sin}(t)\, dt$$

$$= \frac{\pi}{\omega_0}\left[\frac{1}{4} - \frac{1}{4}\right] \times 4 = \underline{0} \quad \text{(similar to prev.)}$$

- $\boxed{q_1^{cos}(t) \,\&\, q_2^{sin}(t)}$ are orthogonal over $[0,T]$;

$$<q_1^{cos}(t), q_2^{sin}(t)> = \int_0^{2\pi/\omega_0} q_1^{cos}(t)\, q_2^{sin}(t)\, dt$$

$$= \frac{\pi}{\omega_0}\left[\frac{1}{2} + \frac{1}{2} - \frac{1}{2} - \frac{1}{2}\right] = \underline{0}$$

- $\boxed{q_2^{cos}(t) \,\&\, q_1^{sin}(t)}$ are orthogonal over $[0,T]$;

$$<q_2^{cos}(t), q_1^{sin}(t)> = \int_0^{2\pi/\omega_0} q_2^{cos}(t)\, q_1^{sin}(t)\, dt$$

$$= \frac{\pi}{\omega_0}\left[\frac{1}{4} - \frac{1}{2} + \frac{1}{4} - \frac{1}{4} + \frac{1}{2} - \frac{1}{4}\right] = \underline{0}$$

c) $x(t) = a_0 + \sum_{k=1}^{\infty} a_k q_k^{cos}(t) + \sum_{k=1}^{\infty} b_k q_k^{sin}(t)$

- Taking inner-product with $q_m^{cos}(t)$ on both sides,

$\langle x(t), q_m^{cos}(t)\rangle = \cancel{\langle a_0, q_m^{cos}(t)\rangle} \to 0$ ($\because$ full-cycle & H.W.S.)

$+ \sum_{k=1}^{\infty} \langle a_k q_k^{cos}(t), q_m^{cos}(t)\rangle + \sum_{k=1}^{\infty} \cancel{\langle b_k q_k^{sin}(t), q_m^{cos}(t)\rangle} \to 0$

$\Rightarrow \langle x(t), q_m^{cos}(t)\rangle = a_k \int_{\langle T\rangle} \cancel{q_k^{cos}(t) q_m^{cos}(t)\, dt} \to 1$, $k=m$ ($\because 0\ \forall k \neq m$)

$\Rightarrow \boxed{a_k = \frac{1}{T}\int_{\langle T\rangle} x(t) q_k^{cos}(t)\, dt}$

- Taking inner-product with $q_m^{sin}(t)$ on both sides,

$\langle x(t), q_m^{sin}(t)\rangle = \cancel{\langle a_0, q_m^{sin}(t)\rangle} \to 0$

$+ \sum_{k=1}^{\infty} \cancel{\langle a_k q_k^{cos}(t), q_m^{sin}(t)\rangle} \to 0 + \sum_{k=1}^{\infty} \langle b_k q_k^{sin}(t), q_m^{sin}(t)\rangle$

$\Rightarrow \langle x(t), q_m^{sin}(t)\rangle = b_k \int_{\langle T\rangle} q_k^{sin}(t) q_m^{sin}(t)\, dt$, $k=m$

$\Rightarrow \boxed{b_k = \frac{1}{T}\int_{\langle T\rangle} x(t) q_k^{sin}(t)\, dt}$

- Taking inner-product with 1 on both sides

$\langle x(t), 1\rangle = \langle a_0, 1\rangle + \sum_{k=1}^{\infty} \cancel{\langle a_k q_k^{cos}(t), 1\rangle} \to 0 + \cancel{\sum_{k=1}^{\infty} \langle b_k \ldots\rangle} \to 0$

$\Rightarrow \boxed{a_0 = \frac{1}{T}\int_{\langle T\rangle} x(t)\, dt}$

d) $x(t) = \begin{cases} 1 & , \ 0 \le t \le 1 \\ 0 & , \ 1 < t \le 2 \end{cases}$

$T_x = 2.$

$\Rightarrow a_k = \frac{1}{2}\int_0^2 x(t)\, q_k^{cos}(t)\, dt$

$= \underbrace{\frac{1}{2}\int_0^1 q_k^{cos}(t)\, dt}$ → (effective area of plot from $t = 0 \to 1$)

$\Rightarrow b_k = \overbrace{\frac{1}{2}\int_0^1 q_k^{sin}(t)\, dt}$

$\Rightarrow a_0 = \frac{1}{2}\int_0^2 x(t)\, dt = \underline{\underline{1/2}}$

e) Consider $q_{k_1}^{cos}(t)$ and $q_{k_2}^{cos}(t)$.

$\Rightarrow \langle q_{k_1}^{cos}(t), q_{k_2}^{cos}(t) \rangle = \int_{\langle T \rangle} q_{k_1}^{cos}(t)\, q_{k_2}^{cos}(t)\, dt$

Analytical argument: The two signals' ~~k values (k_1, k_2) are submultiples of~~ time periods are submultiples of the original $T = 2\pi/\omega_0$. By considering half the cycle of the signal with $k_1 > k_2$, the inner product obtained complements the inner product from the later half cycle. Thus, the resultant is zero, and the signals are orthogonal ($k_1 \neq k_2$).